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Terraform Variables:-
- ⇒ Reusable.
- → 3 types:-
  - i) Primitive
  - ii) Collection
  - iii) Structural.
- ↳ Can be in different file formats: like
  - least priority ⟷ variables.tf
  - Most priority ⟷ terraform.tfvars
  - ↓
  - variables.tf
  - is needed compulsory.

Outputs:-

→ To get output of arguments after creation instead of searching in .tf state.

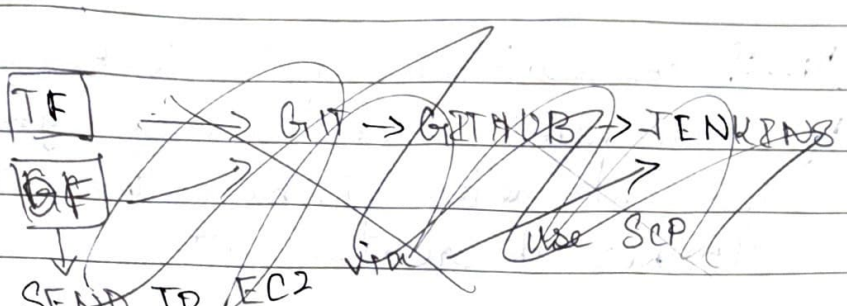
ssh -i 'Private key pair file path' ubuntu@54.242.

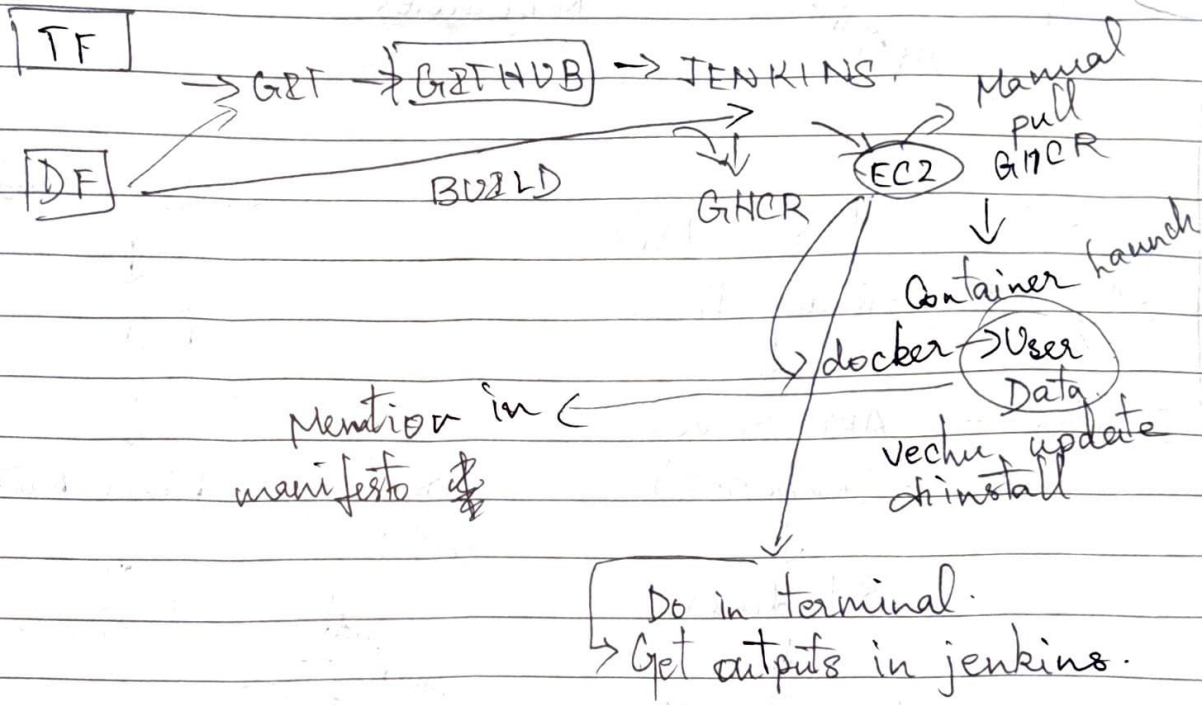
↳ To directly connect to instance in terminal.

Project:-
- → Write terraform Script
- → Push Script from git to github
- → Pull to Jenkins & automate it.
  - ↳ Plan, Validate.
- → Inside the instance, install docker
  - ↳ Using user data
  - ↓
  - Build container.

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→ Git → GitHub.
→ Write Docker Compose file
→ After EC2, get public IP and transfer Through Jenkins, from local to user data (EC2) + build it.





Kubernetes:-

→ Orchestration tool

Orchestrates docker (Commands). → Takes care of fault tolerance.

→ Direct communication with Kubernetes.

→ Eg:- You want 3 containers, you tell Kubernetes, Kubernetes tells docker & creates it.

*Collection of containers is a (Pod) → Smallest deployable unit

Kubernetes.

Master Node (Control Plane)

Gathers requests ↑ frontend ← API Server (KubeAPI)

↓ Scheduler

↓ Controller-Manager (Maintaines decide state:- Actual State)

Replica

↓ ETCD (Key-Value Database)

Worker Node

↓ Pod

↓ docker / containerd ← Container Runtime

↓ updates work to master ← Kubelet

↓ Kube proxy & Sharing responsibility (Kai maathi vidrathu)

Controller - Manager Compones :-

1) Node controller.
2) Deployment controller
3) Replica - set controller (Pod. Health check)
4) Job controllers. (Kudutha Job a check pannum)

Kubernetes is an open-source container management tool which automates container deployment, container (de) scaling & container load balancing.

↳ Robust & Reliable.
↳ Best soln. for scaling up containers.